

CSC 212: Data Structures and Abstractions

01: Introduction

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CSC 212

Learn to model and solve complex
problems with computers.

Course description

- Introduction to **fundamental data structures** and their algorithms
 - ✓ arrays, lists, stacks, queues, trees, hash tables, graphs (most popular topics for job interview questions)
 - ✓ survey of classic algorithms for sorting and searching
- Basic principles of analysis of algorithms
 - ✓ improve your foundation of CS theory
- Writing code that runs efficiently
 - ✓ choosing good algorithms and data structures

Presumes a strong
foundation in C++
programming, including:

- Pointers
- Classes and objects
- Recursion

Course information

- Course Website
 - ✓ Syllabus, Schedule, Resources
 - ✓ <https://homepage.cs.uri.edu/~malvarez/teaching/csc-212/>

What is expected from you?

- Attend lectures/labs
 - ✓ students are expected to attend all lectures and labs
 - ✓ regular attendance is linked to higher grades and better comprehension of course material
- Participate and think critically
 - ✓ ask questions (lectures, labs, office hours, Ed, ...)
- Start working on assignments early
 - ✓ refrain from simply copying and pasting answers generated by LLMs
- Laptops and cellphones are **NOT permitted** unless being used for taking notes

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Need a refresher on C++ programming?

- Pick a textbook (learn syntax)
- Solve Challenges



Kattis



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Warming up

- Adjacent elements sum
 - ✓ find the **maximum sum** of any pair of adjacent elements in an array of integers

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11	3	5	3	2	-5	6	7	-9	-2	13	2
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first pair

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Practice

- Complete the following program:

```
#include <vector>
#include <iostream>

int max_sum(std::vector<int>& nums) {
    // TODO
    return -1;
}

int main() {
    std::vector<int> nums = {1, -2, 3, 4, -1, 2, 1, -5, 4};
    int result = max_sum(nums);
    std::cout << "Maximum sum is: " << result << std::endl;

    return 0;
}
```

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